Autumn 2021 Su

MATH 126 Midterm Exam 2
Tuesday November 23, in your quiz section

Name: Quiz section:

7-digit UW ID:

Exam Instruction:

- You have 50 minutes to complete the 5–page exam. Distribute your time wisely.
- Show your work to earn full credit.
- **Do NOT write within 1 cm of the edge!** Your exam will be scanned for grading.
- Leave your answer in the exact form rather than a decimal approximation. For example, you may leave your answer as a fraction, a square root expression, an inverse trig function of an expression, etc.
- You can prepare one hand-written double-sided 8.5" × 11"–inch page of notes and bring it to the exam.
- You may use a basic calculator that can not graph, differentiate, or integrate (e.g., TI-30X IIS). **All other electronic devices (e.g., cell phone, earbuds) are forbidden.**
- You must finish the exam independently. **Giving or receiving any assistance on the exam is considered cheating, which will result in a grade of zero for the exam.** There are multiple versions of the exam, do not discuss the exam questions with other students on the exam day.
- **Please tear off the last page of scratch paper. Do NOT turn in the scratch paper unless you have put work on it to be graded.**

1. (12 pts) Let $z = f(x, y)$ be a function defined by the implicit equation 1(a) 14.3

$$xz + e^z = y^2$$

(a) Use implicit differentiation to find $\dfrac{\partial z}{\partial x}$ and $\dfrac{\partial z}{\partial y}$.

1(b) 14.4

(b) Use your answer in (a) to give a vector tangent to the trace of the surface $z = f(x, y)$ on the plane $x = 2$ at the point $(2, 1, 0)$. (The question did not ask for the tangent line, just give a tangent vector.)

2. (15 pts) Consider the function $f(x, y) = x^2 + \frac{1}{3}y^3 - 3y$.

2 14.7

(a) Find all the critical points of f in $\mathbb{R}^2$, then determine if f has a local maximum, local minimum, or a saddle point at each of the critical point. (No need to find the f value at the critical points for this part.)

(b) Find the absolute maximum and absolute minimum value of f on the disk $D = \{(x, y) \,|\, x^2 + y^2 \leq 1\}$.

3. (12 pts) Consider the integral

$$\int_0^1 \int_{2x}^2 \cos(y^2)\, dydx$$

(a) Sketch and shade the region of integration on the xy–plane.

(b) Reverse the order of integration and evaluate the integral.

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Chapter 15 material - covered in Math 224

4. (10 pts) Setup a double integral in **Polar coordinate** to find the volume of the solid below the cone $z = 6 - \sqrt{x^2 + y^2}$ and above the paraboloid $z = x^2 + y^2$. Do **NOT** evaluate the integral.

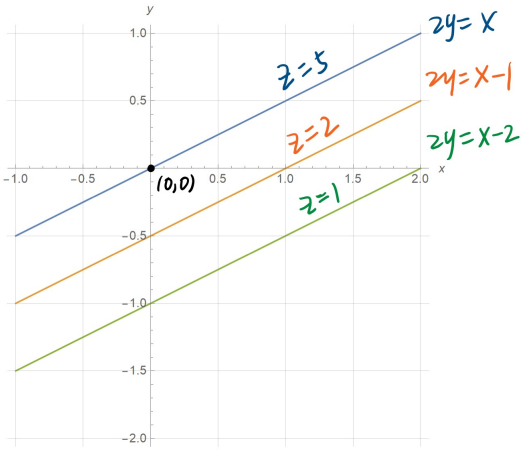
$$\int_{\square}^{\square} \int_{\square}^{\square} \square \, d\square \, d\square$$

To get full and/or partial credit, show work to arrive the upper/lower limits of the integral.

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5. (11 pts) The graph below illustrated 3 level curves of an unknown function $z = f(x, y)$. 5(a,b) 14.1/14.3
lines are: $2y = x$ with $z = 5$; $2y = x - 1$ with $z = 2$; $2y = x - 2$ with $z = 1$.



(a) Give a reasonable (and as precise as possible) estimation for $f_x(0, 0)$ and $f_y(0, 0)$.

(b) Based on the given level curves, would you say the second derivative $f_{xx}(0, 0)$ is positive or negative? Briefly explain your answer.

(c) (**This is a challenge question that worth 3 points, do not spend much time on this question unless you have confidently done the other problems.**) 5(c) 14.1

Find a function $z = f(x, y)$ that has the given three level curves. The answer is not unique.

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